

GIT tutorial

Pauline Hubert and Nadia Lafrenière

Sage Days 130.5, May 5, 2026

based on a talk given with Pauline Hubert at Sage Days 105

Motivations

"FINAL".doc



FINAL.doc!



FINAL_rev.2.doc



FINAL_rev.6.COMMENTS.doc



FINAL_rev.8.comments5.
CORRECTIONS.doc



FINAL_rev.18.comments7.
corrections9.MORE.30.doc



FINAL_rev.22.comments49.
corrections.10.#@\$%WHYDID
ICOMETOGRADSCHOOL?????.doc

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What software to use? Pros and cons

	Pros	Cons
Email	"Easy" to use	So many!
Overleaf	Many people can work at the same time	Only online, latex only
Dropbox	Easy to use	No two people can work at the same time, no version control
Git	Version control, automatic fusion of modifications, any file type	Learning curve

What is Git?

Git is a *version control software*.

- ▶ It keeps track of every modifications.
- ▶ You can go back in time to a previous version.
- ▶ It allows many people to work in parallel.

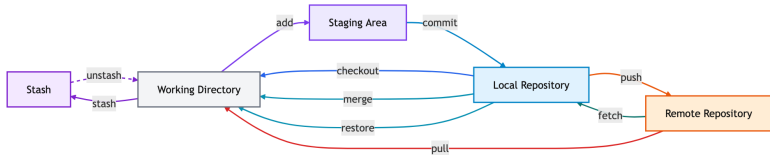
Online servers for Git

Store your data somewhere.

- ▶ Local: on your computer.
- ▶ Online servers: GitHub, Bitbucket, GitLab, your university server, etc.

If you use an online server, you will also have a local version on your computer. If you want to work with other people, you have to use a server.

Workflow



Picture: Iannick Gagnon

Now that you are convinced... Installation

- ▶ Works on Windows, Mac or Linux
- ▶ Go to <https://git-scm.com/install/> or (Unix systems)
`sudo apt install git`

To share code, you need to identify yourself:

```
$ git config --global user.name "John Doe"
```

```
$ git config --global user.email "john.doe@email.com"
```

Please, put your real name instead.

To get code from a remote repository:

```
$ git clone https://address-of-the-repo.git
```

Maybe some of you did, yesterday,

```
nadia@xps:~/software$ git clone git://github.com/sagemath/sage.git
Cloning into 'sage'...
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
Receiving objects: 4% (25444/636096), 4.81 MiB | 3.20 MiB/s
```

What's happening?

status

Lost? You wanna check if everything went well? This command is your friend!

```
$ git status
```

```
nadia@xps:~/Documents/Exposés mathématiques/2019/git_sagedays2019$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git checkout -- <file>..." to discard changes in working directory)

        modified:   git-tutorial.tex

no changes added to commit (use "git add" and/or "git commit -a")
```

First steps: Starting to use Git on a project

`init`

Two ways to start:

- ▶ Use an existing local project.
- ▶ Create a new project on a server (e.g. GitHub) and clone it.

If you already have a project.

1. In the directory containing your files, run the command `git init`.
2. (optional) Then, you can put your files on the server. For that, create a new *empty* repository on the server and follow the instructions.

The commands you will need are

```
$ git remote add origin REPO_URL
```

```
$ git push -u origin master
```

Once you worked on your project, you want to save the modifications.

To select the (new or modified) files that will be added to the modification use the command

```
$ git add FILE_NAME1 FILE_NAME2 ...
```

To save modifications, use the command `git commit`.

Each time you make a commit, a text editor appears in your terminal and you have to write a short (but *meaningful*) comment to describe your modifications.

If you modified only files that were already in the git repository you can also list the files directly in the commit command.

```
$ git commit FILE_NAME1 FILE_NAME2 ...
```

Two usefull options : `git commit -OPTS`, where `-OPTS` is

- `-a` allows you to save all the modified files with the same commit message (as long as you initially added them).
- `-m "Comment"` allows you to directly write your commit message in the command.

For the moment, all your modifications are only saved in your local copy of your project.

After committing, you may want to share your modifications by adding them to the online repository. The command for that is

```
$ git push
```

You may be asked to enter your username and password at this point.

You don't have to push every time you commit. You can push several commits at once, but your collaborators will not see the changes until you push.

Getting updated code & solving conflicts

pull

To obtain the updated version of the project, use

```
$ git pull
```

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“What if one of my collaborator worked at the same time as me?”

- ▶ If you did not work on the same part of the project (e.g. the same line or paragraph), all edits go through.
- ▶ If you worked on the same line, you create a conflict that you'll have to fix it manually. Follow the instructions.

Reviewing changes

diff

You can look at what other people did:

► on GitHub

The screenshot shows a GitHub pull request interface. At the top, there are tabs for Conversation (6), Commits (3), Checks (26), and Files changed (2). The pull request title is "Enhance exponent handling in numeric::pow_intexp to support exact Python objects and improve type checks". The commit hash is 55195a9, and it was committed by cxzhong last week. The diff shows changes to the file src/sage/functions/log.py. The changes include adding a new function class Function_exp(GinacFunction) and updating the log function to handle exact Python objects. The diff highlights the changes with a green background. The file src/sage/symbolic/ginac/numeric.cpp is also shown with changes.

Conversation 6 Commits 3 Checks 26 Files changed 2 +20

Commit 55195a9 Submit review

Filter files...

src/sage

functions

log.py

symbolic/ginac

numeric.cpp

Enhance exponent handling in numeric::pow_intexp to support exact Python objects and improve type checks

cxzhong committed last week commit 55195a9

src/sage/functions/log.py

```
@@ -158,6 +158,14 @@ class Function_exp(GinacFunction):
158 sage: 2*sqrt(e) # needs sage.symbolic
159 2*e^(1/2)
160
161 + Test substitution into an exponential whose argument contains ``I``,
162 + During simplification the coefficient of the logarithm can be an exact
163 + Python object equal to the integer ``-1``:
164 +
165 + sage: t, q = var('t, q') # needs sage.symbolic
166 + sage: exp(-I*pi*q).subs(q=-1/pi*log(2*t)) # needs sage.symbolic
167 + 1/2/t
168 +
169 Check that :issue:`19918` is fixed:
170
171 sage: exp(-x^2).subs(x=oo) # needs sage.symbolic
```

src/sage/symbolic/ginac/numeric.cpp

You can look at what other people did:

► on your files

```

Soit  $G$  un groupe fini et soit  $S$  un  $G$ -module simple. Soit  $\chi$  le caractère de  $S$ . Alors, on
note  $\rho_S$  l'élément de  $\mathbb{C}G$ 
-    $\rho_S = \frac{1}{\dim(S)} \sum_{g \in G} \overline{\chi(g)} g$ 
+    $\rho_S = \frac{1}{\dim(S)} \sum_{g \in G} \overline{\chi(g)} g$ 
+   où  $\overline{\chi}$  désigne le conjugué complexe de  $\chi$ .
La multiplication à droite par cet élément constitue la fonction sur l'algèbre du groupe  $G$ 
\begin{align*}
\text{isoproj}_S : \mathbb{C}G &\rightarrow \mathbb{C}G^S \\
&w \mapsto w \cdot \rho_S, \\
&w \mapsto w \cdot \rho_S.
\end{align*}
-   où  $\overline{\chi}$  désigne le conjugué complexe de  $\chi$ .

```

Reviewing changes

diff

To know what has been change since last commit:

```
$ git diff
```

or since commit1

```
$ git diff commit1
```

or between commit1 and commit2

```
$ git diff commit1 commit2
```

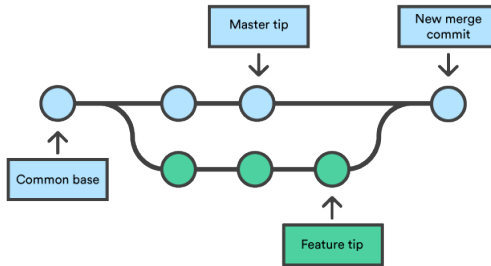
To recover the name of commit1, you type `git log` (this is the history of the repository). The name is very long! (e.g.: `df14e25763073c18e30ebee3d34ef466dd08c8`)

diff



Branches

Image from Bitbucket web site

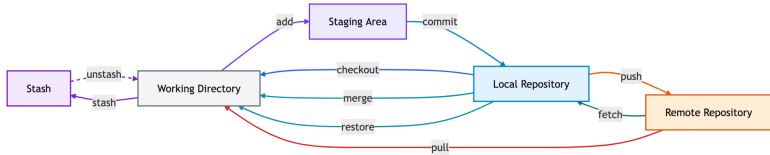


Create a new branch : `$ git branch BRANCH_NAME`

Navigate between branches : `$ git checkout BRANCH_NAME`

Merge branches : `$ git merge BRANCH_NAME`

Workflow



Picture: Iannick Gagnon

Takeaway

PLEASE! Read the error messages and outputs.

Takeaway

PLEASE! Read the error messages and outputs.

Some references



An awesome tutorial:

swcarpentry.github.io/git-novice/



The tutorial given last week at the Sage Days has extensive documentation, in French



Some cheatsheets, by Software Carpentry and GitHub



Bitbucket, Github and Gitlab all have extended documentation on their website